

MAT 528 HW11

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Problem 1 (Munkres 53.3). Let $p: E \rightarrow B$ be a covering map; let B be connected. Show that if $p^{-1}(b_0)$ has k elements for some $b_0 \in B$, then $p^{-1}(b)$ has k elements for every $b \in B$.

Solution. Given $b \in B$, let $U \subseteq B$ be a neighborhood of b that is evenly covered by p . The slices of $p^{-1}(U)$ are in bijection with $p^{-1}(b)$ by definition, so, for each $b' \in U$, $p^{-1}(b')$ has the same cardinality as $p^{-1}(b)$ (with bijection between the two given by sending $x \in p^{-1}(b)$ to the $y \in p^{-1}(b')$ that is in the same sheet of $p^{-1}(U)$ as x). This means that the cardinality of $p^{-1}(b)$ is a locally constant function of b . Locally constant functions on connected spaces are constant, so the cardinality of $p^{-1}(b)$ is equal to that of $p^{-1}(b_0)$ for all $b \in B$, thus $p^{-1}(b)$ has k elements, too. $\square$

Problem 2 (Munkres 53.5). Show that the map $p: S^1 \rightarrow S^1$ given by $p(z) = z^n$ is a covering map.

Solution. Let $z = \cos \theta + i \sin \theta \in S^1$ for some real θ . Then,

$$p^{-1}(z) = \left\{ \cos \left(\frac{\theta + 2\pi k}{n} \right) + i \sin \left(\frac{\theta + 2\pi k}{n} \right) : k = 0, 1, \dots, n-1 \right\}.$$

If $\varepsilon = \frac{\pi}{100}$, then the neighborhood $U = \{\cos \alpha + i \sin \alpha : |\theta - \alpha| < \varepsilon\}$ has pre-image

$$p^{-1}(U) = \bigcup_{k=0}^{n-1} \left\{ \cos \alpha + i \sin \alpha : \left| \frac{\theta}{n} - \alpha \right| < \frac{\varepsilon}{n} \right\}.$$

Let the argument of the union be U_k ; since ε is sufficiently small, the U_k are pairwise disjoint. We show that $p|_{U_k}: U_k \rightarrow U$ is a homeomorphism. By construction, this restricted map is bijective and continuous, and its inverse is continuous because $p|_{U_k}$ is an open map — in fact, the unrestricted $p: S^1 \rightarrow S^1$ is already an open map, since it sends $\{\cos \alpha + i \sin \alpha : \alpha \in (a, b)\}$ for some $a, b \in \mathbb{R}$ to the open set $\{\cos \alpha + i \sin \alpha : \alpha \in (na, nb)\}$, and sets of this kind generate the topology of S^1 .

The work above shows that p is a local homeomorphism $S^1 \rightarrow S^1$, and of course p is continuous and surjective (the latter of which can be seen by e.g. noting that $p^{-1}(z)$, which we described above, is nonempty for all $z \in S^1$). $\square$

Problem 3 (Munkres 54.5). Consider the covering map $p: \mathbb{R} \times \mathbb{R} \rightarrow S^1 \times S^1$ of Example 4 of §53. Consider the path $f(t) = (\cos 2\pi t, \sin 2\pi t) \times (\cos 4\pi t, \sin 4\pi t)$ in $S^1 \times S^1$. Sketch what f looks like when $S^1 \times S^1$ is identified with the donut surface D . Find a lifting $\bar{f}$ of f into $\mathbb{R} \times \mathbb{R}$, and sketch it.

Problem 4 (Munkres 54.6). Consider the maps $g, h: S^1 \rightarrow S^1$ given by $g(z) = z^n$ and $h(z) = 1/z^n$. Compute the induced homomorphisms g_* and h_* of the infinite cyclic group $\pi_1(S^1, b_0)$ into itself.

Solution. Since S^1 is path connected, we can, without loss of generality, fix $b_0 = 1$ (since the change-of-basepoint homomorphism is an isomorphism). Let $\alpha: [0, 1] \rightarrow S^1$ be an arbitrary loop in S^1 based at 1 and let $p: \mathbb{R} \rightarrow S^1$ be the standard covering map. Suppose α lifts to a path $\tilde{\alpha}$ in $\mathbb{R}$ starting at 0; then, $g \circ \alpha$ lifts to the path $n \cdot \tilde{\alpha}$ because, for all $t \in [0, 1]$,

$$p(n \cdot \tilde{\alpha}(t)) = \cos(n \cdot \tilde{\alpha}(t)) + i \sin(n \cdot \tilde{\alpha}(t)) = (\cos(\tilde{\alpha}(t)) + i \sin(\tilde{\alpha}(t)))^n = \alpha(t)^n = g(\alpha(t)),$$

where the second-to-last equality is by definition of $\tilde{\alpha}$ being a lift of α . Letting $\phi: \pi_1(S^1, b_0) \rightarrow \mathbb{Z}$ be the lifting correspondence used in Munkres's Theorem 54.5 (which is an isomorphism), we see that

$$\phi([g \circ \alpha]) = n\phi([\alpha]) = \phi(n[\alpha]),$$

pulling n into ϕ because ϕ is a homomorphism; and since ϕ is an isomorphism, applying its inverse shows that $\boxed{g_*([\alpha]) = [g \circ \alpha] = n[\alpha]}$.

Similarly, $h(z) = 1/z^n = z^{-n}$ is pretty much the same map — using $-n$ in place of n above shows that $\boxed{h_*([\alpha]) = -n[\alpha]}$ for all $[\alpha] \in \pi_1(S^1, b_0)$. $\square$

Problem 5. Generalize the proof of Theorem 54.5 to show that the fundamental group of the torus is isomorphic to the group $\mathbb{Z} \times \mathbb{Z}$.

Solution. Let $p: \mathbb{R} \times \mathbb{R} \rightarrow S^1 \times S^1$ be the covering map in Munkres's §53, Example 4, let $e_0 = (0, 0)$, and let $b_0 = p(e_0)$. Then $p^{-1}(b_0)$ is the set $\mathbb{Z} \times \mathbb{Z}$ of ordered pairs of integers. Since $\mathbb{R}^2$ is simply connected, the lifting correspondence

$$\phi: \pi_1(S^1 \times S^1, b_0) \rightarrow \mathbb{Z} \times \mathbb{Z}$$

is bijective. We show that ϕ is a homomorphism (where $\mathbb{Z}^2$ has the componentwise additive structure), and the theorem is proved.

Given $[f]$ and $[g]$ in $\pi_1(S^1 \times S^1, b_0)$, let $\tilde{f}$ and $\tilde{g}$ be their respective liftings to paths in $\mathbb{R}^2$ beginning at $(0, 0)$. Let $(a, b) = \tilde{f}(1)$ and $(c, d) = \tilde{g}(1)$; then $\phi([f]) = (a, b)$ and $\phi([g]) = (c, d)$, by definition. Let $\tilde{\tilde{g}}$ be the path

$$\tilde{\tilde{g}}(s) = (a, b) + \tilde{g}(s)$$

in $\mathbb{R}^2$. Because $p((a, b) + x) = p(x)$ for all $x \in \mathbb{R}$, the path $\tilde{\tilde{g}}$ is a lifting of g ; it begins at (a, b) . Then the product $\tilde{f} * \tilde{\tilde{g}}$ is defined, and it is the lifting of $f * g$ that begins at $(0, 0)$. The endpoint of this path is $\tilde{\tilde{g}} = (a, b) + (c, d)$, i.e.

$$\phi([f][g]) = (a, b) + (c, d) = \phi([f]) + \phi([g]).$$

$\square$

Problem 6 (Munkres 55.4). Suppose you are given the fact that for each n , there is no retraction $r: B^{n+1} \rightarrow S^n$. Prove the following:

- (a) The identity map $i: S^n \rightarrow S^n$ is not nullhomotopic.
- (b) The inclusion map $j: S^n \rightarrow \mathbb{R}^{n+1} - \{0\}$ is not nullhomotopic.
- (c) Every nonvanishing vector field on B^{n+1} points directly outward at some point of S^n , and directly inward at some point of S^n .
- (d) Every continuous map $f: B^{n+1} \rightarrow B^{n+1}$ has a fixed point.
- (e) Every $(n+1) \times (n+1)$ matrix with positive real entries has a positive eigenvalue.
- (f) If $h: S^n \rightarrow S^n$ is nullhomotopic, then h has a fixed point and h maps some point x to its antipode $-x$.

Solution. (a) Suppose i is nullhomotopic, and let $H: S^n \times I \rightarrow S^n$ be a homotopy from i to a constant map. We prove a generalization of (1) $\implies$ (2) in Lemma 55.3. Let $\pi: S^n \times I \rightarrow B^{n+1}$ be the map $\pi(x, t) = (1-t)x$. This is continuous, closed, and surjective, so it is a quotient map, collapsing $S^1 \times \{1\}$ to the point 0 , and is otherwise injective. Since H is constant on $S^1 \times \{1\}$, it induces an extension $B^{n+1} \rightarrow S^n$ of i via the quotient map π ; this extension is exactly a retraction because it must be constant on S^n , in accordance with i . We are given that no retraction $B^{n+1} \rightarrow S^n$ exists, which is a contradiction, so i is not nullhomotopic.

(b) Let $r: \mathbb{R}^{n+1} - \{0\} \rightarrow S^n$ be the retraction $r(v) = v/|v|$; then $i = r \circ j$ is the identity map, so if j is homotopic to some constant map $c: S^n \rightarrow \mathbb{R}^{n+1}$, then i is homotopic to the constant map $r \circ c$, which contracts part (a).

(c) Associate to each $x \in B^{n+1}$ the vector $v(x)$. Suppose first that $v(x)$ does not point directly inward at x for any $x \in S^n$. Let w be the restriction of v to S^n . Noting that B^{n+1} is the quotient of $S^n \times I$ under the quotient map π from part (a), we find that $v \circ \pi: S^n \times I \rightarrow \mathbb{R}^n$ is a homotopy of the map w to a constant map since $v \circ \pi$ is constant on the fiber of the point $0 \in B^{n+1}$. That is, w is nullhomotopic.

On the other hand, w is homotopic to the inclusion map $j: S^n \rightarrow \mathbb{R}^{n+1} - \{0\}$ via the homotopy $F(x, t) = tx + (1-t)w(x)$; to verify that $F(x, t)$ is nowhere zero, we note that $F(x, t) = 0$ is equivalent to $w(x)$ being a negative scalar multiple of x , which we have assumed doesn't hold. It follows that j is nullhomotopic, which contradicts the result of (b).

The analogous result for an outward-pointing vector follows from applying the work above to the vector field $(x, -v(x))$.

(d) Suppose that $f(x) \neq x$ for every $x \in B^{n+1}$. Then, defining $v(x) = f(x) - x$ gives us a nonvanishing vector field $(x, v(x))$ on B^{n+1} . But the vector field cannot point directly outward at any point x of S^1 , as that would mean $f(x) - x = ax$ for some positive real a , so that $f(x) = (1+a)x$ would lie outside B^{n+1} . We thus arrive at a contradiction.

(e) Let A be the subset of S^n with all nonnegative coordinates; A is easily seen to be homeomorphic to the ball B^n . Then, treating the given matrix as a linear transformation T with respect to the standard basis on $\mathbb{R}^n$, the map $x \mapsto T(x)/|T(x)|$ has a fixed point x_0 by part (d), implying $T(x_0) = |T(x_0)|x_0$, so that $|T(x_0)|$ is a positive eigenvalue for T .

(f) As proven in part (a), since h is nullhomotopic, it has an extension $\tilde{h}: B^{n+1} \rightarrow S^n$. Since $0 \notin S^n$, the vector field $(x, \tilde{h}(x))$ on B^{n+1} is nonvanishing, so it points directly outward at some point of S^n and directly inward at some point of S^n . The former corresponds to a fixed point of $\tilde{h}$ and the latter to a point x that gets mapped by $\tilde{h}$ to $-x$ because $\tilde{h}$ takes on values in S^n only. $\square$